



Two psychometric profiles of gifted children in referred samples: The need to distinguish between homogeneous and crystallized profiles

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ABSTRACT

Detecting giftedness is a high-stake issue in psychological assessment. Often defined as intellectual performance in the top 2.5 %, giftedness concerns a substantial share of the general population; giftedness identification is also one of the most frequent motivations for intelligence testing in children. Despite the importance of this topic, there has rarely been consistent research in the methods used to classify an individual as gifted in the context of psychological assessment. Psychometric profiles associated with giftedness, in particular, have not been systematically explored, despite the possibility that there are subtypes of giftedness with different implications for research and practice. We collected the assessment results of a sample of 739 children identified as gifted after completing Wechsler's Intelligence Scale for Children (WISC-IV or WISC-V), in one of five different testing locations in France or Belgium. Profile analyses revealed the existence of two distinct psychometric profiles, each comprising roughly half the sample. One group of children had consistently high performance on all indices; whereas the other group had very high performance specifically on verbal comprehension subtests tapping into crystallized intelligence, but mild performance everywhere else. These results clearly support the idea that there are two roads to giftedness: some children reach the cutoff of giftedness based on high cognitive abilities overall, whereas others are particularly adept at leveraging a capital of acquired knowledge in verbal tasks. These two profiles have different implications for the real-life achievement of gifted individuals, leading to major consequences for both research and clinical practice.

1. Introduction

Gifted individuals have the potential for becoming high achievers and making meaningful contributions to society (Tannenbaum, 1997). At the same time, gifted children have special educational needs and benefit greatly from specific instructional practices (such as acceleration, ability grouping or enrichment: see Rogers, 2007; Steenbergen-Hu et al., 2016). This makes the accurate identification of giftedness a high-stake issue in psychological assessment – and most definitions of giftedness are based on high intellectual ability. While contemporary conceptions of giftedness tend to emphasize talent and exceptional abilities as emerging from multiple factors, conceptual models of giftedness consider high intelligence either as synonymous with the term

“giftedness” (as opposed to talent, which may or may not develop in gifted individuals; Gagné, 2004), or as a key component or precursor of giftedness (Heller et al., 2005; Renzulli, 1999; Sternberg, 2005).

The importance of intelligence to giftedness is reflected in testing strategies: the most common approach for identification of giftedness relies on intelligence testing (Cao et al., 2017; Carman, 2013; McClain & Pfeiffer, 2012; Ziegler & Raul, 2000; although academic achievement often also contributes to identification: for an overview, see McClain & Pfeiffer, 2012; Subotnik et al., 2011) – usually with the criterion of performing in the top 2.5 % of intellectual ability, corresponding to a full-scale IQ (FSIQ) at least 130. Having a robust understanding of the typical patterns of performance of gifted individuals on intelligence tests is therefore critical, both to research (the quality of research on

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giftedness is contingent on the quality of sample selection), and to practice (identification of giftedness has become one of the leading reasons to seek intelligence testing, especially in children).

A major difficulty is that there is no indication that giftedness is a unitary, self-consistent label and that gifted individuals form a homogeneous group. On the contrary, a substantial literature has suggested that the performance of gifted individuals in intelligence tests is highly heterogeneous, and relatedly, that there are multiple ways to be “gifted” – with prior data from research and applied settings pointing to a frequent discrepancy between high reasoning ability and high ability to acquire and leverage knowledge (e.g. Gonthier, 2025; for references, see the supplemental materials on <https://osf.io/c87hw/>).

Surprisingly, there has never been a systematic test of the possibility that there exist separable profiles of performance on intelligence tests for gifted individuals. Substantial heterogeneity in gifted samples may reflect different psychometric profiles, indicating the existence of different subtypes of giftedness with different implications for research and practice. Testing this possibility was the objective of the current study. In the next sections, we review the literature regarding heterogeneity in gifted samples, before turning to profile analysis with the Wechsler scales.

1.1. Giftedness and heterogeneity in intellectual performance

Intelligence is not a completely homogeneous ability. Although all intelligence tests tend to correlate and can be summarized with a general factor (g), this general factor encompasses multiple broad abilities (see the CHC taxonomy of intelligence: Schneider & McGrew, 2018). Intelligence tests usually assess a mix of fluid intelligence (Gf: logical reasoning), visual processing (Gv), crystallized intelligence (Gc: declarative knowledge), working memory (Gwm) and processing speed (Gs). Subtests assessing different broad abilities are affected by different variables: for example, crystallized intelligence tasks depend on both verbal ability and acquired knowledge, and are much more affected by cultural capital and socio-economic level than other abilities (e.g. Georgas et al., 2003; Gonthier et al., 2021). This can create inconsistencies in test performance, with a given individual performing higher in one area than other. In turn, differences between broad abilities may be particularly relevant to understanding giftedness and tailoring solutions to each specific individual (Gridley et al., 2003; Warne, 2015).

Samples of gifted individuals have, on average, higher scatter of performance between the various aspects of intelligence than the non-gifted, with sometimes very large discrepancies between indices. This is true both for cognitive ability profiles in general (e.g. Lohman et al., 2008), and for Wechsler sales in particular (Gonthier, 2025). This heterogeneity in gifted individuals is sometimes mistaken for a pathological profile and an instance of “developmental asynchrony” or “dysharmonious development”¹ (e.g. Guénolé et al., 2015), learning disability (Habib, 2023), or even brain damage (Waldron & Saphire, 1990). In fact, however, heterogeneity in gifted individuals actually appears unrelated to negative outcomes in large cohorts (Shevchenko et al., 2023), and constitutes a normal and expected phenomenon.

Indeed, the correlation between scores on different types of intelligence tests decreases when ability increases (for a meta-analysis, see Blum & Holling, 2017; see also Breit, Brunner, et al., 2022): in other words, individuals with high ability tend to have more heterogeneous scores on different indices of performance, and more frequent discrepancies (e.g. Grégoire et al., 2011). This is an instance of Spearman’s law

of diminishing returns (Spearman, 1927), also called ability differentiation. A possible cause is that a low ability comes with a bottleneck on cognitive processes that creates all-around low performance, whereas a high ability affords more room for marked strengths and weaknesses (Detterman & Daniel, 1989; Kovacs & Conway, 2016). As a result, gifted individuals tend not to have perfectly flat, homogeneous profiles of performance (although some datasets may give this impression due to ceiling effects on certain intelligence tests: Achter et al., 1996, Achter et al., 1997; Milgram & Hong, 1999).

1.2. Discrepancies in patterns of performance on the wechsler scales for gifted individuals

Heterogeneity in gifted performance is particularly manifest in the Wechsler scales, with which most research has been conducted (see Gonthier, 2025). Wechsler scales assess crystallized intelligence (Verbal Comprehension Index [VCI]), fluid intelligence and visuo-spatial processing (confounded in a Perceptual Reasoning Index [PRI] representing logical reasoning on abstract visuo-spatial materials in earlier versions, and broken down into a Fluid Reasoning Index [FRI] and Visuo-Spatial Index [VSI] in more recent versions), working memory (Working Memory Index [WMI]), and processing speed (Processing Speed Index [PSI]). The average performance of gifted individuals on Wechsler scales shows substantial disparities between these indices, with the typical pattern: VCI > PRI > WMI > PSI. This pattern is ubiquitous in the literature (as summarized in Gonthier, 2025; see also the supplemental materials: <https://osf.io/c87hw/>) and has been reported since the first version of the WISC (Gallagher & Lucito, 1961).

This typical pattern of VCI > PRI > WMI > PSI shows several things. First, gifted individuals tend to score substantially lower (although still above average) on the WMI and PSI than on the PRI and VCI. Some researchers have proposed that this is due to gifted subjects adopting a more deliberate approach, with lower response speed and less errors, which could penalize them on time-sensitive subtests (Kaufman, 1992; Reams et al., 1990; Volker & Smerbeck, 2009). On the other hand, this could also be a simple statistical artifact: working memory and processing speed are related to general intelligence, but usually to a lower extent than crystallized and especially fluid intelligence (e.g. Reynolds et al., 2013). This means in a sample of participants selected for their high general ability, these indices will have a stronger tendency to regress to the mean. Regardless of the causal mechanism, lower scores in these areas are expected, and this has even led some investigators to recommend that giftedness assessment use a corrected full-scale score, the General Ability Index (GAI), which does not include working memory and processing speed subtests (Prifitera et al., 1998; Silverman, 2009; Watkins et al., 2022; but see Rowe et al., 2010, 2012).

On the other hand, the discrepancy frequently observed between the PRI and VCI is more surprising. This discrepancy is highly common in gifted samples (for a meta-analysis, see Gonthier, 2025; these estimates are also summarized in the supplemental materials for the current study on <https://osf.io/c87hw/>). Averaged over all gifted children in a meta-analysis, the average VCI is about 6 points greater than the PRI, where the expected value would be zero; when considering idiographic patterns, the mean absolute discrepancy between VCI and PRI in a given child is closer to 14 points on average. In studies reporting the frequency of discrepancies, it is typical for one-third to two-thirds of gifted children to have a discrepancy of at least 15 points between the two indices – in almost all cases, in the direction VCI > PRI.

This anomaly is also reflected in factor analyses of the Wechsler scales. Factor structure in gifted samples has long been a preoccupation of the field, and appears to be relatively similar to non-gifted samples at the subtest level. However, confirmatory factor analyses also show that latent variables representing the PRI and VCI are poorly subsumed by a general factor of intelligence in gifted samples (see Gonthier, 2025; see also <https://osf.io/c87hw/>). Among six samples where g-loadings were available (Macmann et al., 1991; Molinero et al., 2015; Pezzuti et al.,

¹ These notions are frequently applied to discrepancies between cognitive and socio-emotional development, but they can also be taken to encompass discrepancies between cognitive abilities (this was the case in the first works on the concept of dyssynchrony: Terrassier, 1985). For a discussion, see Silverman, 1997.

2022; Rowe et al., 2014; Watkins et al., 2002; see also Hagmann-von Arx et al., 2008; Liratni & Pry, 2012; Rowe et al., 2010), the VCI and PRI had opposite signs in three cases (i.e., one loaded positively and the other negatively on the general factor); a common *g* factor could not be estimated due to lack of model convergence in one case, and the last two cases had the PRI *g*-loading at 0.97 and the VCI *g*-loading at 0.20 and 0.16. Examining the relation between VCI and PRI more directly leads to the same conclusion: among the eight gifted samples where the relation between VCI and PRI was tested, all samples showed a correlation below 0.30; it was below 0.20 in six cases and negative in two cases. Note that restriction in the range of scores (mechanically found in gifted samples selected to have high performance) is insufficient to account for these findings, given the prevalence of null or negative correlations and the small size of positive relations.

In sum, subtests assessing fluid and crystallized intelligence appear somewhat unrelated – or even mildly opposing – in gifted samples. In other words, the data indicate that it tends to be different gifted individuals who succeed at these different tasks. A divergence between two indices thought to reflect fluid and crystallized intelligence is very unexpected, given that these two abilities usually have two of the highest loadings on the general factor of intelligence (often both in the 0.70–0.90 range: e.g., Carroll, 1993; Reynolds et al., 2013; Schneider & McGrew, 2018), implying that they should provide consistent scores. This finding is broadly compatible with an extreme form of ability differentiation (Breit, Brunner, et al., 2022), but seems to go even further, in that differentiation would not lead to predicting a *negative* correlation between indices as has been repeatedly observed. Severe inconsistency between PRI and VCI scores therefore appears to be a defining feature of gifted samples.

1.3. Two possible profiles of giftedness

The finding that some children have high scores on subtests involving logical reasoning on abstract visuo-spatial reasoning materials, whereas others have selectively high scores on subtests involving the ability to leverage prior knowledge in verbal subtests, may appear surprising given current theories of intelligence which describe these two abilities as highly related aspects of the general factor of intelligence. However, this pattern is compatible with the phenomenon of differentiation (Blum & Holling, 2017; Breit, Brunner, et al., 2022), and can easily be reconciled when considering that these two indices could reflect two different profiles of performance. Indeed, average imbalance between VCI and PRI, frequent large discrepancies at the individual level, and poor second-order factor structure are all highly compatible with the presence of separable clusters of participants with different strengths (although it would be possible to get this pattern without separable profiles, and vice versa).

Although strongly related, fluid and crystallized intelligence have partly different correlates and rely on partly different causal mechanisms. Fluid intelligence is more strongly related to other cognitive abilities such as working memory and processing speed, whereas crystallized intelligence primarily relies on “the depth and breadth of both declarative and procedural knowledge”, as developed “through experience, learning and acculturation” (Schneider & McGrew, 2018). For this reason, the possibility that they represent separable profiles of ability ties in with the distinction, prevalent in the field of giftedness, between intellectual performance and academic achievement. Major authors in the field have presented this in terms of *psychological* versus *educational* definitions of giftedness (Gallagher & Courtright, 1986), or *intellectually gifted* versus *academically talented* (Gagné, 2004; the distinction made by Renzulli, 1999 between *creative-productive* and *schoolhouse* giftedness also has similarities with this idea).

The first definition, and the most commonly used (Carman, 2013), emphasizes performance on psychometric tests of intelligence and logical reasoning: individuals with exceptional cognitive performance (usually defined as FSIQ of 130 or above, but sometimes extended to

exceptional performance for any broad ability: Gridley et al., 2003; McIntosh et al., 2018; Warne, 2015) are labeled as gifted. The second definition emphasizes academic achievement: students with high grades, and/or nominated by teachers as high achievers, are labeled as gifted. This definition is used not only in practice (to some large extent: McClain & Pfeiffer, 2012, cite identification based exclusively on intelligence tests as a “historical relic”), but also in many published studies, partly due to the fact that it is easier to identify large samples of individuals with high academic performance than samples of individuals with high FSIQ (Tourreix et al., 2023).

Of course, these two views of giftedness are not incompatible: intelligence is an excellent predictor of academic achievement (Laidra et al., 2007; Primi et al., 2010), schooling increases reasoning abilities (Ritchie & Tucker-Drob, 2018), and nothing prevents a given child from having both high cognitive abilities and high knowledge. Some authors also combine both aspects to target “intellectually gifted and academically talented pupils” (Gagné, 1995). However, many psychologists seem to implicitly expect that identification strategies based on cognitive abilities and declarative knowledge are interchangeable, and will ultimately lead to the same children, which is not necessarily true in practice. Knowledge and achievement are highly dependent on variables other than intelligence, such as cultural background and socio-economic status, and a hard-working child in the right conditions may achieve excellent knowledge with only average intelligence. Conversely, a child with superior reasoning skills may be bored at school, achieve mediocre grades, fail to be nominated by teachers as an excellent pupil, and display knowledge in the average range. As a result, a child may come to be flagged by parents or teachers for giftedness identification because they display high cognitive abilities, high knowledge, or possibly – but not necessarily – both. High performance on fluid intelligence subtests and high performance on crystallized intelligence subtests therefore constitute two different possible paths to identification as being gifted, and two possible profiles for individuals who meet the criterion for inclusion in a gifted sample.

This interpretation is compatible with prior works on ability profiles. In particular, different patterns of performance in intelligence tests for gifted individuals are stable over time, suggesting that heterogeneity does not merely represent error variance (Breit et al., 2021; Breit, Brunner, et al., 2022; Lohman et al., 2008). Patterns of performance in standardized achievement tests such as the SAT or ACT differentially predict future success and life outcomes not only in non-gifted samples (see prior work regarding ability tilt in the NLSY data, e.g. Coyle et al., 2014; for a review, see Coyle, 2018), but also in gifted samples (for an overview, see Lubinski, 2016), as prominently found in the SMPY longitudinal study of high-ability youths (e.g. Lubinski et al., 2001; see also Achter et al., 1999; Benbow & Minor, 1990; Park et al., 2007; Webb et al., 2007; for a different example with anxiety, see Kermarrec et al., 2020). Academically underachieving gifted students also perform highest on visuo-spatial reasoning subtests (Lupart & Pyryt, 1996), and gifted students with high performance on visuo-spatial reasoning subtests tend to demonstrate substantially lower academic success than gifted students with high performance on crystallized intelligence subtests (Gohm et al., 1998; Lakin & Wai, 2020), to the point where one author presented high ability for visuo-spatial reasoning as an “inaptitude for academic learning” (Lohman, 2005). These studies support the possibility that the “gifted” label in fact encompasses meaningfully different subgroups with different abilities.

1.4. Rationale for the study

As reviewed in the preceding sections, gifted samples are characterized by unexpectedly large heterogeneity in ability scores, an heterogeneity that seems relatively stable over time and that seems to have predictive value. CHC theory does include second-order factors such as fluid and crystallized intelligence that depend on partly different determinants, but these factors are usually very consistent, with extremely

high correlations. This is not what occurs in gifted samples, where *Gf* and *Gc* tend to be nearly orthogonal or even negatively correlated, also showing very large mean discrepancies. While this is broadly consistent with the phenomenon of differentiation at higher ability levels, no theoretical account of giftedness or intelligence would predict such an extent of discrepancy. One possible reason that could reconcile this finding with the literature could be that the gifted are not an homogeneous group, and that individuals can be referred for giftedness testing based either on high cognitive abilities or a high knowledge base – broadly matching the views of giftedness as intellectual ability or academic talent, both prevalent in the literature. If gifted samples can include both individuals with high intellectual abilities and/or individuals with high knowledge, with either being sufficient for identification, this could lead to the observed discrepancy between factors representing *Gf* and *Gc*.

This can be easily tested by examining whether separate psychometric profiles exist in intellectual assessments of gifted individuals. Surprisingly, there has never been a systematic investigation of psychometric profiles of gifted individuals in the Wechsler scales, despite profile analyses with Wechsler scales having drawn considerable interest from clinical psychologists (e.g. Pfeiffer et al., 2000). The purpose of this study was therefore to examine profiles of performance in a sample of gifted children, by way of cluster analyses.

Given prior literature, we expected to find two separable psychometric profiles: one with high cognitive abilities overall, and one with high crystallized intelligence in particular. The group with high cognitive abilities overall was expected to display high scores primarily on the PRI (for the WISC-IV) or FRI/VSI (for the WISC-V), and to a lower extent on the WMI and PSI which tend to correlate rather well with fluid intelligence but to a lower extent than other second-order factors. The group with high crystallized intelligence in particular was expected to display high scores primarily on the VCI, which is especially dependent on declarative knowledge. Given our hypothesis that either high cognitive abilities or high declarative knowledge are sufficient to be referred for giftedness testing, and given the phenomenon of regression to the mean (which makes it difficult to obtain very high scores on multiple variables), we expected the first group to score relatively lower on the VCI and the second group to score relatively lower on the PRI/FRI/VSI/WMI/PSI, although still above average due to the correlation of all these variables with the general factor of intelligence.

Our objective was to help characterize the profiles of children likely to be encountered by psychologists; therefore, we based our study on the test results of children referred to psychologists and being tested for giftedness identification, then subsequently identified as gifted. To ensure that the results were generalizable, we recruited the sample across five testing locations in two different countries, and we included children tested with either the WISC-IV or WISC-V. Studies in the field of giftedness often include very small samples (Tourreix et al., 2023); here we strived to collect data from several hundreds of participants to ensure the stability of our results.

Given the often limited stability of cluster analyses when using different analytic strategies, we also ensured that the results were replicable with different methods. Local giftedness identification practices in our countries use either the criterion of FSIQ at least 130, or the criterion of any index at least 130 (Gridley et al., 2003; Warne, 2015); we first conducted our analyses on the broader sample of children with any index at least 130 to maximize sample size, but we also replicated them on the more selective subsample of children with FSIQ at least 130, to ensure that the finding of heterogeneous profiles could not be due to the use of multiple inclusion criteria.

Three complementary analyses were also conducted to help interpret the results. First, we needed to ensure that the results were generalizable (e.g. over testing locations), and that they were not driven by one particular subgroup in the sample. This was done by testing the relation between profiles and socio-demographic variables: age, gender, and testing location. Second, we needed to ensure that the results were

specific to children being tested for giftedness identification (rather than an artifact of the analytic methods, or the scale composition). This was done by replicating the profile analyses in the standardization samples for the WISC-IV and WISC-V. Third, we needed to ensure that separate profiles found in the current dataset actually reflected a differentiation between indices of performance (rather than a discrepancy in absolute values between still highly-correlated indices). This was done by testing whether the intended factor structure of the Wechsler scales applied in our dataset using confirmatory factor analysis (CFA) and examining model fit and covariance between indices; we also compared this factor structure directly to the standardization sample using a test of measurement invariance.

2. Method

This research received approval n°10022023 from the [Nantes University] institutional review board.

2.1. Transparency and openness

The study design and analysis were not formally preregistered. We do not have authorization to share the data (collected for counselling or clinical purposes and potentially identifying) or research materials (the Wechsler scales). Sample R code for all data analyses is available on <https://osf.io/c87hw/>. Packages used for analysis are detailed in the next sections.

2.2. Participants

Data were collected from five different locations specialized in the identification of gifted children: four in France, and one in French-speaking Belgium. All patients were tested specifically for intelligence assessment (with typical motives such as “boredom at school” or “exceptional performance”) at their or their parents’ demand. All tests were conducted by trained psychologists, with at minimum a master’s degree in psychology, and with specific expertise in giftedness assessment. Testing locations comprised a mix of psychologists with individual offices and larger-scale assessment services, private or public. We are confident that this represents a fairly diverse sample of the more common testing locations for giftedness available to patients in French-speaking countries (school psychologists and non-specialized mental health care centers also perform intelligence tests, but more rarely in the context of giftedness assessment).

All available patient records were retrieved and included in the study, provided that they met four criteria: 1) tested with either the WISC-IV (Wechsler, 2003) or WISC-V (Wechsler, 2014); 2) within the normal testing conditions for the WISC (e.g., age comprised between 6 years old and 16 years and 11 months old); 3) received a score superior or equal to 130 on at least one index of the WISC; and 4) no condition that could interfere with performance (e.g. non-fluent French speaker, major visual or auditory disability, medical or psychiatric condition...). Other possible confounding factors, such as the presence of autism spectrum disorder, could not be systematically checked, but the corresponding records were excluded when the information was available. Note that identifying children as “gifted” based on a score superior or equal to 130 was the practice in all testing locations included here (although the formal definition of giftedness varied from one psychologist to another).

In total, 739 children were included in the study. This number included 659 children tested with the WISC-IV (502 boys and 157 girls; mean age = 10.00 years old, *SD* = 2.92, range = 6 to 16.95), and a smaller sample of 80 children tested with the WISC-V (58 boys and 22 girls; mean age = 9.91 years old, *SD* = 2.93, range = 6 to 16.67). Out of the 739 children in the study, 423 children (57 %) met the more stringent criterion of having a full-scale IQ above 130 (including *n* = 386 for the WISC-IV and *n* = 37 for the WISC-V). Other information such as

family income or education level was not usually available in the records and was coded differently by different psychologists, precluding systematic analysis; based on available information, most children appeared to be from middle to upper class families with middle to high education levels. No information was collected regarding ethnicity.

For comparison purposes, we also retrieved the datasets from the French standardization samples of the WISC-IV ($n = 1103$) and WISC-V ($n = 1049$) from the publisher. These samples were collected by psychologists hired by the publisher (on a paid-by-protocol basis) during the process of developing the WISC. All psychologists received specific training from the publisher prior to data collection; each psychologist sent back protocols to the publisher to ensure that they complied with data collection instructions and that the test was scored correctly. Both samples were collected using quota sampling so as to be approximately representative of the French population in terms of sex, age, geographical region, and parental education.

2.3. Materials and data processing

The WISC-IV, published in France in 2005, includes 10 main subtests each belonging to one of four index scores: the Verbal Comprehension Index (VCI, reflecting the crystallized intelligence factor G_c in the CHC model); the Perceptual Reasoning Index (PRI, reflecting a mix of fluid intelligence factor G_f and visual processing factor G_v); the Working Memory Index (WMI, reflecting the short-term and working memory factor G_{wm}); and the Processing Speed Index (PSI, reflecting the processing speed factor G_s). The WISC-V, published in 2014, also includes 10 main subtests, with a composition similar to the WISC-IV except that the PRI is subdivided in a Fluid Reasoning Index (FRI, reflecting the fluid intelligence factor G_f) and a Visuo-Spatial Index (VSI, reflecting the visual processing factor G_v).

The main subtests and corresponding indices are summarized in Table 1. Supplementary subtests available in the WISC were rarely used by the psychologists in our sample, and were not analyzed here. We analyzed performance on the subtests themselves (standard scores with $M = 10$, $SD = 3$), the index scores ($M = 100$, $SD = 15$) and the FSIQ ($M = 100$, $SD = 15$). All these scores are standardized in reference to age, which naturally accounts for age differences in the sample.

We retrieved records from participating locations and transcribed the scores of all children between 6 years old and 16 years 11 months old with at least one index score above 130 on the WISC-IV or WISC-V, as long as the record included standard scores for all 10 main subtests, the index scores, and the FSIQ. We also retrieved the age and gender of all children. Obvious data entry errors (such as scores outside of the possible range, or FSIQ or index scores incompatible with the subtest standard scores) were corrected. A small number of missing subtest scores (47 out of 6600, or 0.71 %) were replaced using multiple imputation (conducted using package *mice* for R: see van Buuren & Groothuis-Oudshoorn, 2011).

2.4. Data analysis

Profile analysis was primarily conducted on subtest standard scores, after standardizing (z-scoring) all measures to put them on the same scale. The analysis was primarily performed in the WISC-IV sample of all children with at least one index superior or equal to 130 ($n = 659$), then replicated in the smaller comparison sample of the WISC-V ($n = 80$), and in the WISC-IV subsample of children with FSIQ superior or equal to 130 ($n = 386$). The subsample of WISC-V children with FSIQ superior or equal to 130 ($n = 37$) was too small to be analyzed separately.

The profile analysis itself involved two steps,² implemented in R (R

Core Team, 2022). The first step was to determine the number of clusters present in the data. Clusters were created with hierarchical agglomerative clustering (computing euclidean distances between all observations, then agglomerating observations using Ward's D2 method). Like deciding the number of factors to retain in exploratory factor analysis, there is no universal cutoff for cluster analyses and different indices usually point to different solutions; here we selected the optimal number of clusters using package *NbClust*, which implements a series of indices available in the literature to determine the best number of clusters. We used the 28 non-graphical indices with 1- to 15-clusters solutions, and we selected the solution agreed upon by the most indices (majority rule: Charrad et al., 2014). Because both subtest scores and index scores are reasonable bases for a profile analysis of the WISC, and to ensure that the results were not driven by a couple of subtests, this analysis was replicated using index scores instead of subtest standard scores.

The second analytic step was to assign each child in the dataset to one of the identified clusters, for interpretation. This was done with the k-means relocation method, initialized with the results of the hierarchical agglomerative clustering performed in the previous step (see Kassambara & Mundt, 2020). The results were then visualized by comparing the average performance of each cluster on each available measure (subtest standard scores, index scores, and FSIQ, on their original scale). We used between-subjects ANOVAs (with cluster as a between-subjects variable) to test for difference between clusters on each measure, and mixed-design ANOVAs (with cluster as a between-subjects variable and measure as a within-subjects variable, with a focus on the interaction between the two) to test whether the various measures differentially depended on cluster. The stability of cluster assignment was checked using the Jaccard index, as computed from a bootstrap with 1000 resamples (package *fpc*: Hennig, 2007).

Three additional analyses of secondary interest were also conducted. Testing whether the identified profiles were not driven by one of the data collection locations (such that the majority of children with one profile came from a given location), or by children from a certain gender or age, was done using logistic regressions, with profile as a dependent variable and location, gender and age as predictors. Testing whether the profiles identified in the present study were specific to the population of children seeking giftedness assessment was done by replicating all cluster analysis on the WISC-IV and WISC-V standardization samples representative of the general population. Lastly, testing whether separate profiles translated into substantial differentiation at the index level was done with a CFA using package *lavaan* (Rosseel, 2012). This analysis was run on the larger WISC-IV sample including all children with at least one index superior or equal to 130, based on a simple factor structure where each subtest loaded on a single latent variable (representing VCI/PRI/WMI/PSI) also allowing for correlation between latent variables. Modification indices and measurement invariance (compared to the WISC-IV standardization sample) were examined using the same package.

3. Results

Descriptive statistics are available in Table 2. All variables had a distribution close to normal with satisfying range. As expected, scores were substantially lower for working memory and processing speed subtests on average, owing either to lower correlations with g (e.g. in the WISC-IV standardization sample, the FSIQ correlates 0.82 with the VCI and 0.61 with the PSI) or to a lesser contribution of these abilities to the selection of potentially gifted children.

3.1. Analysis of psychometric profiles

For all datasets, the majority of indices used to select the optimal number of clusters in the hierarchical agglomerative analysis (Charrad et al., 2014) pointed to the existence of two distinct profiles (see Table 3). Similar results were obtained when replicating the analysis on

² Robustness checks with alternative clustering methods and alternative cluster assignment methods are not systematically reported here but are available from the authors; they provided similar solutions.

Table 1

List of subtests in the WISC-IV and WISC-V.

Subtest	Description	WISC-IV			WISC-V		
		Items	Order	Index	Items	Order	Index
Vocabulary	Provide a definition for vocabulary words	36	6	VCI	29	6	VCI
Similarities	Find similarities between two concepts	23	2	VCI	23	2	VCI
Comprehension	Comprehension of social situations and concepts	21	9	VCI	–	–	–
Block Design	Reproduce an abstract design using colored blocks	14	1	PRI	13	1	VSI
Visual Puzzles	Find pieces that reconstruct a target when assembled	–	–	–	29	8	VSI
Matrix Reasoning	Find the missing piece of a matrix with logical rules	35	8	PRI	32	3	FRI
Figure Weights	Find missing weights to balance a scale	–	–	–	34	7	FRI
Picture Concepts	Find matching pictures based on conceptual categorization	28	4	PRI	–	–	–
Digit Span	Recall digits in forward, backward, or ascending order	32	3	WMI	54	4	WMI
Letter-Number Sequencing	Recall digits and letters in ascending / alphabetical order	30	7	WMI	–	–	–
Picture Span	Recall a series of pictures in serial order	–	–	–	26	9	WMI
Coding	Copy abstract symbols corresponding to numbers	NA	5	PSI	NA	5	PSI
Symbol Search	Determine if target is present in a set of abstract symbols	NA	10	PSI	NA	10	PSI

Note. Items is the total number of items or trials in the subtest (except for the two processing speed subtests which are not divided in discrete trials); Order is the order in which the subtest is performed in the WISC.

Table 2

Descriptive statistics for subtest standard scores, index scores and FSIQ in the WISC-IV and WISC-V (whole samples).

Measure	WISC-IV (n = 659)					WISC-V (n = 80)				
	M	SD	skew	kurtosis	range	M	SD	skew	kurtosis	range
VOC	15.94	2.49	−0.27	−0.94	8–19	15.15	1.98	−0.15	−0.55	11–19
SIM	16.93	2.20	−0.89	0.11	7–19	16.09	1.98	0.01	−1.08	12–19
COM	15.73	2.58	−0.36	−0.64	7–19	–	–	–	–	–
BLD	13.46	2.91	0.09	0.25	5–19	13.05	2.32	0.47	−0.27	9–18
VIP	–	–	–	–	–	13.98	2.01	0.38	−0.09	10–19
MAT	13.83	2.39	−0.41	1.00	4–19	14.26	1.78	0.18	0.10	11–19
FIW	–	–	–	–	–	14.59	2.27	−0.02	−0.50	10–19
PCO	13.33	2.76	0.03	−0.48	7–19	–	–	–	–	–
DSP	12.88	2.78	0.01	0.25	3–19	13.51	2.50	−0.15	0.91	5–19
LNS	12.83	2.55	−0.11	−0.10	5–19	–	–	–	–	–
PSP	–	–	–	–	–	14.07	2.53	−0.88	0.03	8–18
COD	11.41	3.22	0.09	−0.06	3–19	11.96	2.99	0.28	−0.19	6–17
SYM	12.09	2.98	−0.15	0.62	1–19	12.06	2.64	−0.19	−0.87	6–19
VCI	139.25	11.89	−0.57	0.11	90–156	130.64	9.73	−0.13	−0.39	111–151
PRI	124.90	12.36	−0.21	0.04	75–155	–	–	–	–	–
VSI	–	–	–	–	–	119.68	10.91	0.23	−0.23	100–147
FRI	–	–	–	–	–	125.86	9.47	−0.04	0.68	103–155
WMI	117.28	13.21	−0.10	−0.22	73–150	121.19	11.25	−0.26	0.10	94–150
PSI	111.30	15.76	−0.02	−0.31	69–150	111.22	13.66	0.09	−0.52	83–144
FSIQ	133.55	9.49	0.11	−0.39	104–158	129.09	6.55	−0.31	0.23	111–143

Note. VOC = Vocabulary, SIM = Similarities, COM = Comprehension, BLD = Block Design, VIP = Visual Puzzles, MAT = Matrix Reasoning, FIW = Figure Weights, PCO = Picture Concepts, DSP = Digit Span, LNS = Letter-Number Sequencing, PSP = Picture Span, COD = Coding, SYM = Symbol Search.

Table 3

Number of decision indices favoring a given number of profiles.

Type of analysis	Dataset	Number of profiles														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Based on subtest scores	WISC-IV	5	10	4	0	0	1	0	3	0	0	0	0	0	0	5
	WISC-IV with FSIQ $\geq$ 130	4	10	3	2	1	2	0	0	0	0	0	1	1	0	4
	WISC-V	5	9	2	3	0	0	1	0	1	0	0	0	2	0	5
Based on index scores	WISC-IV	5	11	1	0	2	1	1	1	1	1	0	0	0	0	4
	WISC-IV with FSIQ $\geq$ 130	5	9	0	1	5	2	0	0	0	1	0	0	1	2	2
	WISC-V	5	9	4	0	1	0	0	0	1	1	2	1	1	0	3

index scores, with the majority of indices pointing to two clusters as the optimal solution in all cases. We therefore retained a two-profile solution for the remainder of the analyses. The results of the hierarchical agglomerative analysis are displayed in Fig. 1, with the two clusters highlighted: the dendrograms showed a rather clean-cut separation between two profiles.

The pattern of performance on all subtests and indices for the two profiles, after profile assignment using *k*-means, is represented in Fig. 2

(for the analysis based on subtest scores; analyzing the data based on index scores provided practically identical results: see Fig. S1 on <https://osf.io/c87hw/>). In all three datasets, the first profile had a homogeneous profile with high scores on all subtests, despite relatively lower scores on verbal subtests than the second profile (hereafter *homogeneous profile*). The second profile had high scores on verbal subtests representing crystallized intelligence, and relatively lower scores on all other subtests; in other words, these children performed very well specifically

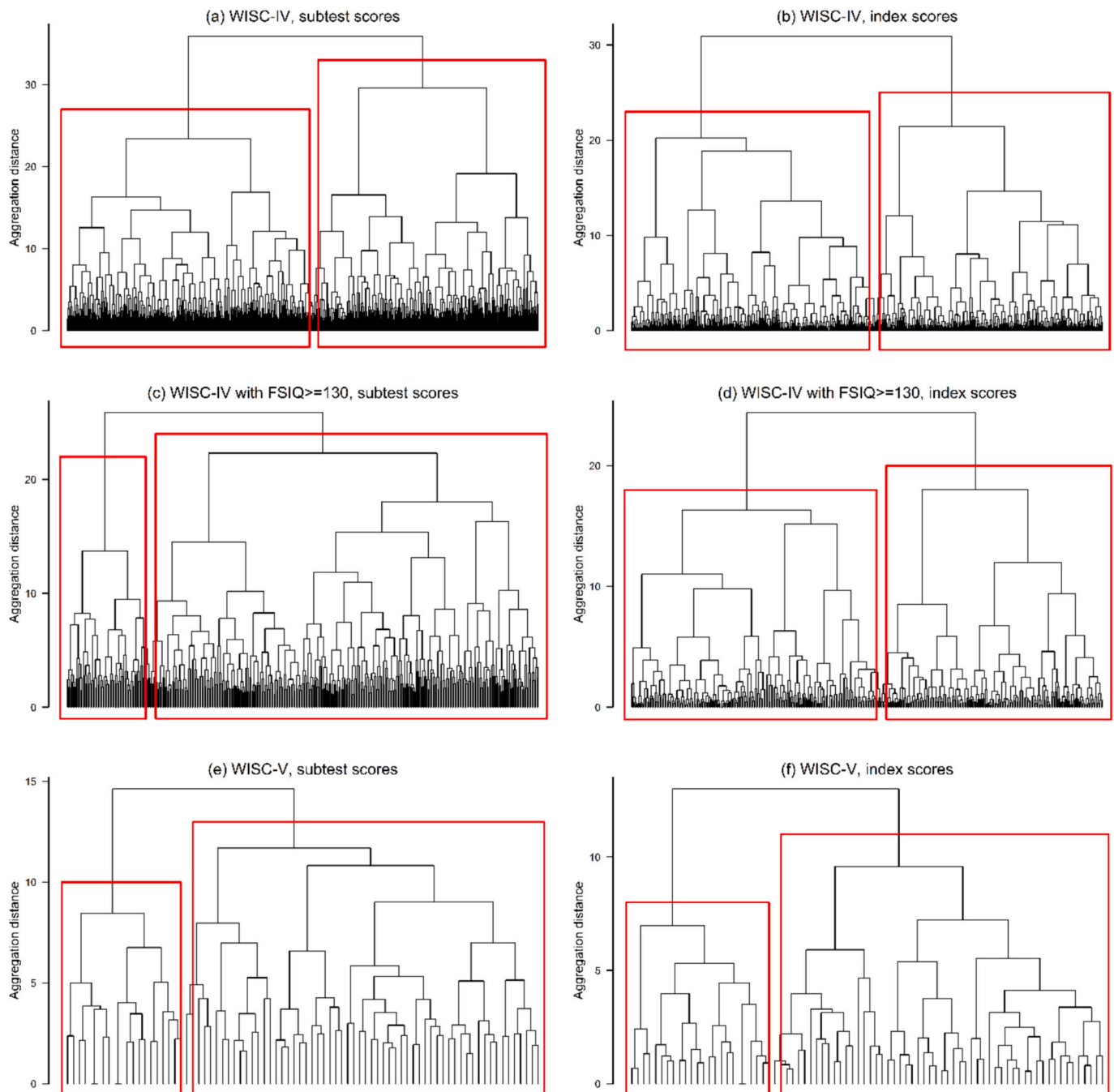


Fig. 1. Dendrograms for the hierarchical agglomerative analysis with a two-cluster solution.

on verbal tasks requiring the use of knowledge (hereafter *crystallized profile*). Children were spread relatively evenly over the two profiles, with a slightly lower proportion of the crystallized profile in the WISC-V (WISC-IV: 52 % crystallized, WISC-IV restricted to FSIQ superior or equal to 130: 59 % crystallized, WISC-V: 41 % crystallized). For all analyses, cluster assignment was reasonably stable (Jaccard indices in the 0.75–0.89 range; Hennig, 2007).

Descriptive statistics and follow-up comparisons are detailed in Table 4, Table 5 and Table 6 for the three datasets. Overall, the two profiles differed on almost all measures in all three datasets (this is expected in cluster analysis, which maximizes the distance between the two profiles). Correcting for multiple comparisons, using Benjamini-Hochberg false discovery rate correction, did not change any of the results. Full-scale IQ was significantly lower in children with a crystallized

profile for the WISC-IV and WISC-V datasets, but not for the WISC-IV dataset restricted to children with FSIQ superior or equal to 130, presumably due to the more restricted range of values.

More importantly, all interactions between profile and measure were significant, indicating differentiated patterns of performance (in other words, the two profiles were not just distinguished by an overall average difference of ability). In terms of index scores, gifted children with a homogeneous profile performed significantly lower on average than children with a crystallized profile on the VCI, but significantly higher on the PRI (WISC-IV), VSI and FRI (WISC-V), WMI, and PSI. In terms of subtests, gifted children with a homogeneous profile performed significantly lower than children with a crystallized profile on all subtests of the VCI, and significantly higher on all other subtests (with the exception of the Matrix Reasoning subtest of the WISC-V where the difference

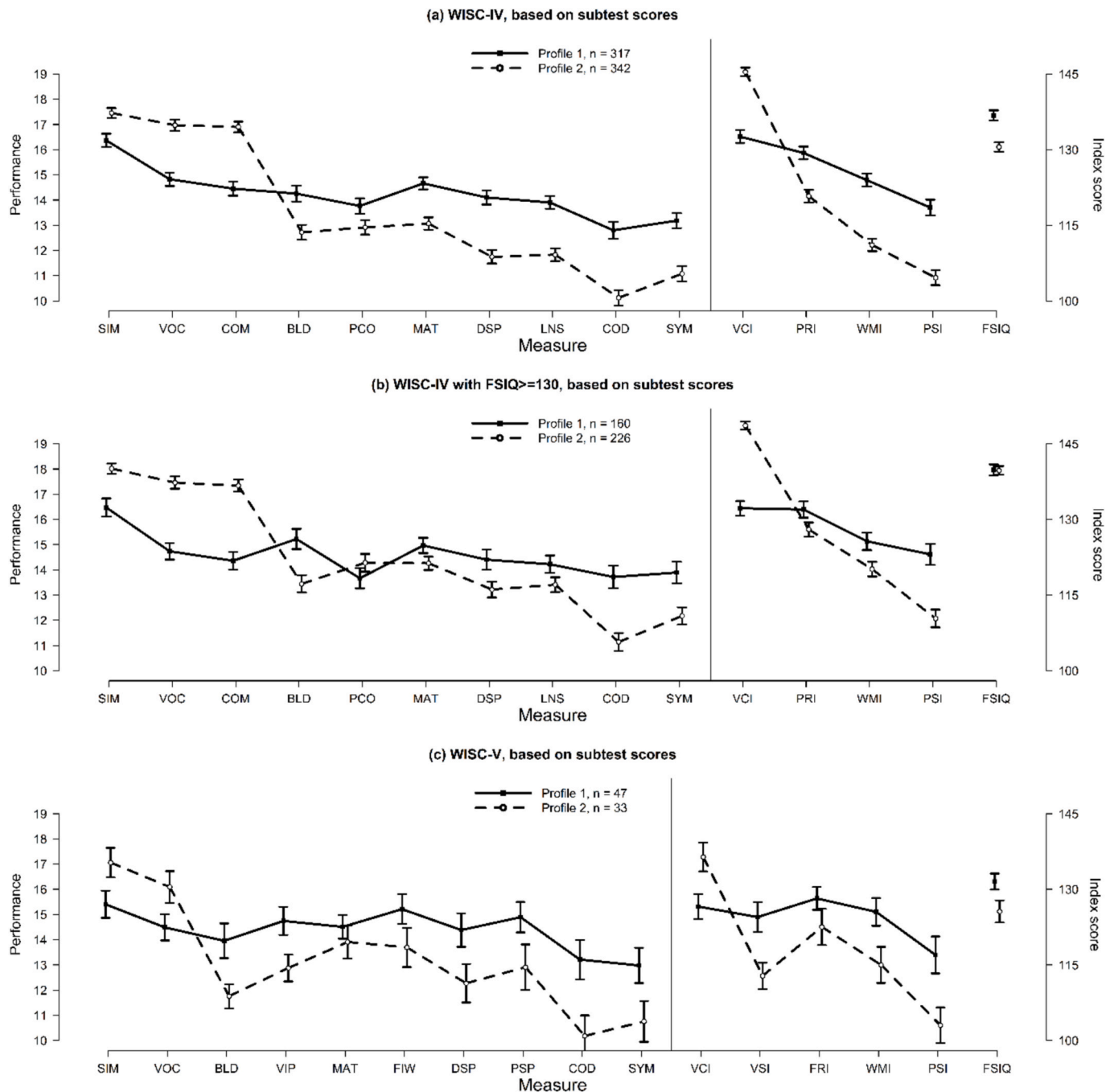


Fig. 2. Mean performance in the two clusters (as computed based on subtest scores) for the three datasets. Errors bars represent 95 % confidence intervals for the mean. VOC = Vocabulary, SIM = Similarities, COM = Comprehension, BLD = Block Design, VIP = Visual Puzzles, MAT = Matrix Reasoning, FIW = Figure Weights, PCO = Picture Concepts, DSP = Digit Span, LNS = Letter-Number Sequencing, PSP = Picture Span, COD = Coding, SYM = Symbol Search.

was not significant).³

Another way to approach these results is to consider the variability of scores within a given profile. In the WISC-IV sample with a FSIQ at least 130, the homogeneous profile had an average VCI of 132.23 and an average PSI of 123.08, a 10-points difference; more generally all indices were within 10 points of each other (see Table 5). By contrast, the

crystallized profile was characterized by a severe discrepancy between scores on the VCI and scores on other indices (see Fig. 2): for instance, in the WISC-IV dataset restricted to children with FSIQ superior or equal to 130, the average VCI was 148.63 for this profile whereas the average PSI was 110.36, a 38-points difference.

A possible criticism of these results could be that the two profiles are a statistical artifact due to averaging, and do not correspond to patterns of performance actually found at the individual level. Examining the scores of individual children showed that this was not the case: in the WISC-IV sample, for example, the VCI was over 30 points above the PRI for 19 % of children, and over 30 points above the average of indices other than the VCI for 36 % of children. The scores of the two children

³ Supplementary subtests were used too rarely by psychologists for a dedicated analysis; only Arithmetic in the WISC-IV had enough data with $n = 34$ and 30 children per profile, and showed a significant advantage for the homogeneous profile, $p < .001$, consistent with its high loading on the Gf and Gwm factors.

Table 4

Descriptive statistics and comparison between the two profiles for the WISC-IV sample.

Measure	Profile 1: Homogeneous		Profile 2: Crystallized		Main effect of profile			Measure * profile interaction
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>F</i>	<i>p</i>	η_p^2	
VOC	14.83	2.40	16.97	2.10	149.45	<0.001	0.19	$F(9, 5913) = 102.92, p < .001, \eta_p^2 = 0.14$
SIM	16.37	2.41	17.46	1.85	42.68	<0.001	0.06	
COM	14.45	2.54	16.90	2.00	190.79	<0.001	0.23	
BLD	14.26	2.85	12.72	2.76	49.16	<0.001	0.07	
MAT	14.66	2.18	13.07	2.32	82.39	<0.001	0.11	
PCO	13.77	2.77	12.92	2.68	15.95	<0.001	0.02	
DSP	14.10	2.55	11.75	2.48	144.24	<0.001	0.18	
LNS	13.91	2.30	11.83	2.37	129.45	<0.001	0.16	
COD	12.80	3.01	10.12	2.85	137.45	<0.001	0.17	
SYM	13.19	2.72	11.08	2.86	93.85	<0.001	0.12	
VCI	132.59	11.74	145.43	8.09	270.35	<0.001	0.29	$F(3, 1971) = 187.49, p < .001, \eta_p^2 = 0.22$
PRI	129.35	11.48	120.78	11.71	89.85	<0.001	0.12	
WMI	123.95	11.61	111.09	11.50	203.88	<0.001	0.24	
PSI	118.52	14.29	104.60	14.02	159.17	<0.001	0.20	
FSIQ	136.80	9.22	130.54	8.74	80.12	<0.001	0.11	–

Note. VOC = Vocabulary, SIM = Similarities, COM = Comprehension, BLD = Block Design, MAT = Matrix Reasoning, PCO = Picture Concepts, DSP = Digit Span, LNS = Letter-Number Sequencing, COD = Coding, SYM = Symbol Search.

Table 5

Descriptive statistics and comparison between the two profiles for the WISC-IV subsample of children with FSIQ superior or equal to 130.

Measure	Profile 1: Homogeneous		Profile 2: Crystallized		Main effect of profile			Measure * profile interaction
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>F</i>	<i>p</i>	η_p^2	
VOC	14.74	2.12	17.46	1.87	178.69	<0.001	0.32	$F(9, 3456) = 65.82, p < .001, \eta_p^2 = 0.15$
SIM	16.48	2.31	18.02	1.53	62.32	<0.001	0.14	
COM	14.36	2.24	17.35	1.88	201.77	<0.001	0.34	
BLD	15.22	2.59	13.45	2.62	43.58	<0.001	0.10	
MAT	14.97	1.94	14.27	2.05	11.55	<0.001	0.03	
PCO	13.67	2.61	14.28	2.68	5.03	0.025	0.01	
DSP	14.41	2.60	13.22	2.42	21.03	<0.001	0.05	
LNS	14.22	2.23	13.42	2.23	12.38	<0.001	0.03	
COD	13.72	2.89	11.14	2.72	79.80	<0.001	0.17	
SYM	13.89	2.76	12.17	2.60	38.90	<0.001	0.09	
VCI	132.23	9.29	148.63	6.25	429.91	<0.001	0.53	$F(3, 1152) = 117.74, p < .001, \eta_p^2 = 0.23$
PRI	131.97	10.41	128.01	10.70	13.14	<0.001	0.03	
WMI	125.67	11.19	120.15	11.07	23.13	<0.001	0.06	
PSI	123.08	13.60	110.36	13.50	82.56	<0.001	0.18	
FSIQ	139.82	7.14	139.75	6.41	0.01	0.919	0.00	–

Note. VOC = Vocabulary, SIM = Similarities, COM = Comprehension, BLD = Block Design, MAT = Matrix Reasoning, PCO = Picture Concepts, DSP = Digit Span, LNS = Letter-Number Sequencing, COD = Coding, SYM = Symbol Search.

Table 6

Descriptive statistics and comparison between the two profiles for the WISC-V sample.

Measure	Profile 1: Homogeneous		Profile 2: Crystallized		Main effect of profile			Measure * profile interaction
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>F</i>	<i>p</i>	η_p^2	
VOC	14.49	1.82	16.09	1.84	14.89	<0.001	0.16	$F(9, 702) = 11.95, p < .001, \eta_p^2 = 0.13$
SIM	15.40	1.88	17.06	1.71	16.13	<0.001	0.17	
BLD	13.96	2.41	11.76	1.39	22.18	<0.001	0.22	
VIP	14.74	1.95	12.88	1.56	20.86	<0.001	0.21	
MAT	14.51	1.64	13.91	1.94	2.24	0.14	0.03	
FIW	15.21	2.06	13.70	2.28	9.58	<0.001	0.11	
DSP	14.38	2.32	12.27	2.24	16.52	<0.001	0.17	
PSP	14.89	2.12	12.91	2.65	13.81	<0.001	0.15	
COD	13.21	2.74	10.18	2.39	26.27	<0.001	0.25	
SYM	12.98	2.44	10.76	2.36	16.47	<0.001	0.17	
VCI	126.57	8.54	136.42	8.36	26.23	<0.001	0.25	$F(4, 312) = 19.10, p < .001, \eta_p^2 = 0.20$
VSI	124.49	10.33	112.82	7.63	30.42	<0.001	0.28	
FRI	128.19	7.96	122.55	10.54	7.45	0.008	0.09	
WMI	125.53	9.68	115.00	10.51	21.38	<0.001	0.22	
PSI	116.98	12.77	103.03	10.39	26.84	<0.001	0.26	
FSIQ	131.53	5.49	125.61	6.43	19.58	<0.001	0.20	–

Note. VOC = Vocabulary, SIM = Similarities, BLD = Block Design, VIP = Visual Puzzles, MAT = Matrix Reasoning, FIW = Figure Weights, DSP = Digit Span, PSP = Picture Span, COD = Coding, SYM = Symbol Search.

most representative of the two profiles (both within the ten closest to the centroid of their respective cluster, while being matched for FSIQ, gender and age range) are presented in Table 7.

An alternative representation of the profiles is given in Fig. 3, in the form of a scatterplot contrasting scores on the VCI with average scores on all other indices. This simplification of the data provided a reasonable approximation of the two profiles. Fig. 3 showed that there was no clear-cut separation, no gap between the two profiles: although there were extreme examples of each (e.g. a child in the bottom right corner, with VCI above 150 and other indices close to 100), there were also children with more intermediate patterns of performance.

The representation of the data in Fig. 3 also makes it possible to extract a simple rule of thumb for profile assignment⁴: as is visible in Fig. 3, for both the WISC-IV and WISC-V datasets, the *k*-means clustering analysis approximately classified gifted children in the crystallized profile if the average of indices other than the VCI was below half the VCI + 50. In other words, a child with VCI = 130 was assigned to the verbal/knowledge profile if the average of the other indices was below $(130/2) + 50 = 115$; a child with VCI = 140 was assigned to the verbal/knowledge profile if the average of the other indices was below 120; and a child with VCI = 150 was assigned to the verbal/knowledge profile if the average of the other indices was below 125.

3.2. Complementary analyses

3.2.1. Predictors of psychometric profile

Logistic regressions (with the homogeneous profile coded as 0 and the crystallized profile coded as 1) showed that age was a predictor of psychometric profile in the WISC-IV dataset ($b = 0.11$, $\chi^2(1) = 17.83$, $p < .001$) and WISC-IV dataset restricted to children with FSIQ superior or equal to 130 ($b = 0.11$, $\chi^2(1) = 8.85$, $p = .003$), but the effect did not reach significance in the WISC-V dataset ($b = 0.14$, $\chi^2(1) = 3.01$, $p = .082$). In all three cases, older children at the time of identification were descriptively more likely to belong to the crystallized profile rather than the homogeneous profile; in other words, average age was lower in the homogeneous group (WISC-IV: $M = 9.51$ years old, WISC-IV restricted to FSIQ ≥ 130 : $M = 9.45$, WISC-V: $M = 9.43$) than in the crystallized group (WISC-IV: $M = 10.46$, WISC-IV restricted to FSIQ ≥ 130 : $M = 10.34$, WISC-V: $M = 10.59$).

Gender was not predictive of psychometric profile in the WISC-IV dataset ($b = -0.05$, $\chi^2(1) = 0.08$, $p = .871$; 23 % girls in both profiles) or in the WISC-IV dataset restricted to children with FSIQ superior or equal to 130 ($b = -0.29$, $\chi^2(1) = 1.35$, $p = .246$; 24 % vs. 19 % girls in the two groups), but it was in the WISC-V dataset ($b = 1.50$, $\chi^2(1) = 7.17$, $p = .007$). For the WISC-V, girls were more likely to belong to the homogeneous profile (38 % girls) than the crystallized profile (12 % girls).

Lastly, the effect of the location where the data were collected was ambiguous. There was no effect of location in the WISC-IV dataset ($\chi^2(4) = 1.94$, $p = .748$), where the proportion of children with a crystallized profile ranged between 46 % and 60 % across locations; there was a significant effect in the WISC-IV dataset restricted to children with FSIQ superior or equal to 130 ($\chi^2(4) = 12.60$, $p = .013$), where the proportion of children with a crystallized profile ranged between 38 % and 80 %; and the effect did not reach significance in the WISC-V dataset ($\chi^2(3) = 7.26$, $p = .064$), where the proportion of children with a crystallized profile ranged between 25 % and 75 %. The variations descriptively observed across samples were in all cases driven by more extreme

⁴ The steps were to predict profile assignment based on the VCI and the average of non-VCI indices; for each possible value of the VCI, extract the cutoff value for the average of non-VCI indices predicting 50 % probability of assignment to each profile; then regress these cutoff values of non-VCI indices on the corresponding VCI values. The resulting regression equation was $0.49 \times \text{VCI} + 51.05$, which gives the same values as the rule of thumb proposed here to within IQ rounding error.

proportions in locations with smaller sample sizes; a Bayesian analysis, less sensitive to the imbalance in group sizes, was firmly in favor of the null in all three datasets ($\text{BF}_{01} = 83,056$, $\text{BF}_{01} = 61$, and $\text{BF}_{01} = 19$, respectively).

3.2.2. Test of replication on the WISC standardization samples

To confirm that the profiles found in the sample of children identified as gifted were specific to this population (rather than an artifact of our analytic approach or the scale composition), we ran the same analyses on the WISC standardization samples. To briefly summarize the results (see Figs. S2 and S3 in the supplemental materials: <https://osf.io/c87hw/>), the two standardization samples appeared highly homogeneous and provided no evidence for the existence of distinct profiles. The analysis for both WISC-IV and WISC-V recommended a solution with three clusters, but this just split the samples into clusters with uniformly high, medium and low performance levels (e.g. for the WISC-IV, the first cluster with $n = 332$ had average subtest scores between 11.69 and 13.47, the second cluster with $n = 524$ had average scores between 9.65 and 10.02, and the third cluster with $n = 247$ had average scores between 6.03 and 8.20; the pattern was identical for the WISC-V). Contrary to the results obtained with the gifted samples, there was no indication of a crossover pattern between the VCI subtests and other subtests. This confirms that the finding of two profiles with a crossover between the VCI and other indices is not an artifact of our analytic approach.

3.2.3. Factor structure of the WISC-IV for gifted children

Our dataset for the WISC-IV (for all children with at least one index superior or equal to 130) was of sufficient size to verify if the factor structure of the scale in gifted children was similar to the standardization sample, or whether there was a low correlation between factors representing the VCI and PRI (as suggested by the differentiation literature), or even a negative correlation (as suggested by the giftedness literature, and by the existence of two separate profiles with a cross-over pattern).

A confirmatory factor analysis on the WISC-IV dataset in the gifted sample showed good fit for a simple four-factor solution allowing for covariance between latent factors, $\chi^2(29) = 63.42$, $p < .001$, CFI = 0.948, TLI = 0.919, GFI = 0.986, RMSEA = 0.042, SRMR = 0.043, confirming that the structure of the WISC-IV was comparable to the general population at the subtest level. All subtests loaded on the expected factor, and modification indices did not reveal meaningful cross-loadings. A full test of measurement invariance (see Putnick & Bornstein, 2016) comparing the gifted sample with the WISC-IV standardization sample is summarized in Table 8. Configural invariance (having the same subtests load on the same factors) was tenable; constraining equality between the two groups for factor loadings (metric invariance), intercepts (scalar invariance), and residuals (residual invariance) all degraded model fit in absolute terms, but the differences were small enough to consider the factor structure as strictly invariant (Cheung & Rensvold, 2002). Even a model with strict residual invariance provided an excellent fit to the data.

The one major difference in factor structure is that in the gifted sample, and contrary to typical results in non-gifted samples, the four factors did not form a single second-order factor representing *g*: the latent variable for verbal comprehension correlated negatively with the latent variables for perceptual reasoning ($r = -0.13$), working memory ($r = -0.05$), and especially processing speed ($r = -0.23$), whereas there were positive correlations between the latent variables for perceptual reasoning and working memory ($r = 0.44$), perceptual reasoning and processing speed ($r = 0.21$), and working memory and processing speed ($r = 0.12$). (Note that the negative correlation between verbal comprehension and perceptual reasoning only occurs in the combined sample including both profiles: the same correlation becomes positive when restricting the analysis to one of the two profiles.)

A CFA model with an explicit second-order general factor (as in e.g.

Table 7
Representative profiles of performance for the two clusters in the WISC-IV.

	SIM	VOC	COM	BLD	PCO	MAT	DSP	LNS	COD	SYM	VCI	PRI	WMI	PSI
Alex	16	14	15	13	16	16	13	15	14	15	130	135	124	128
Ben	19	19	19	13	14	12	13	14	10	14	155	121	121	112

Note. VOC = Vocabulary, SIM = Similarities, COM = Comprehension, BLD = Block Design, MAT = Matrix Reasoning, PCO = Picture Concepts, DSP = Digit Span, LNS = Letter-Number Sequencing, COD = Coding, SYM = Symbol Search. The two profiles are for 6–8 years old boys with FSIQ = 143.

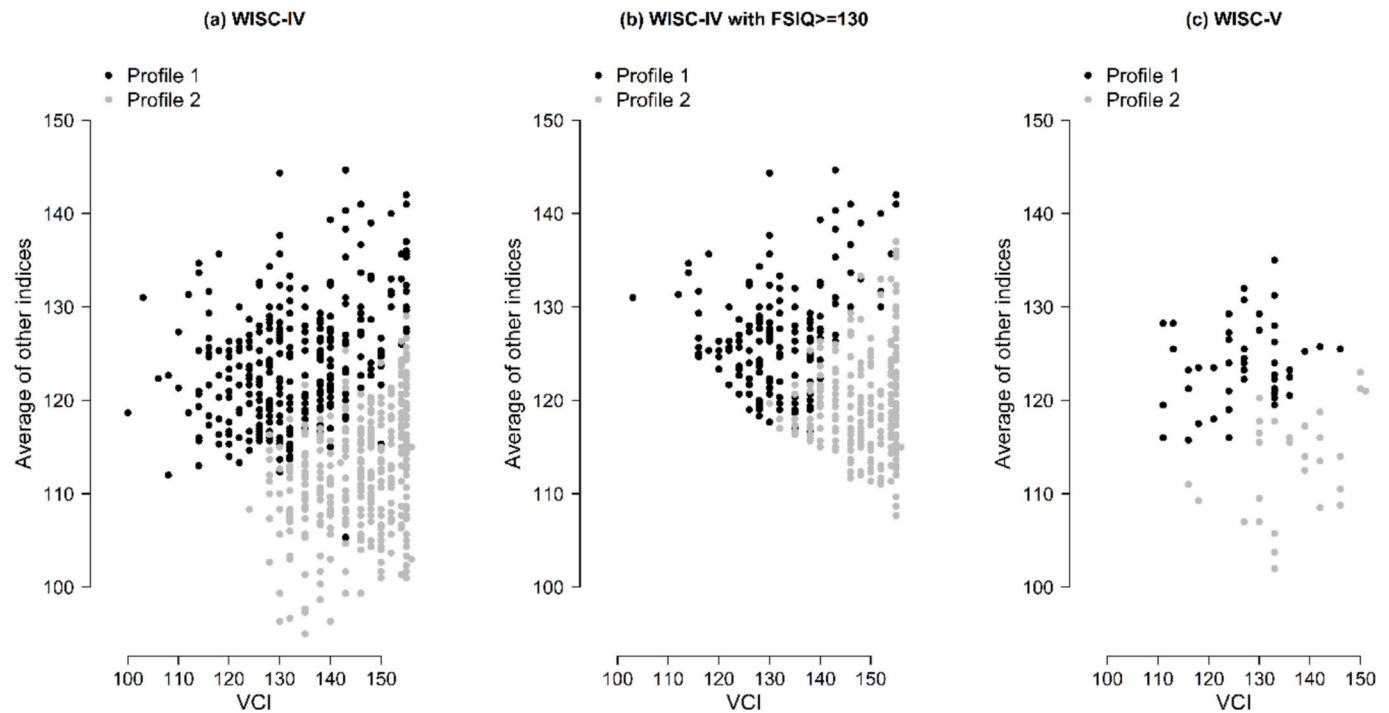


Fig. 3. Scatterplot of all participants for the three datasets. Each point represents a child in the sample, with color indicating the profile assigned by the *k*-means analysis on subtest scores. The x-axis depicts VCI performance and the y-axis depicts the average of the other indices (PRI, WMI and PSI for the WISC-IV datasets and VSI, FRI, WMI and PSI for the WISC-V dataset).

Table 8
Test of measurement invariance between the gifted and standardization samples for the WISC-V.

Model	Absolute fit	CFI	TLI	GFI	RMSEA	SRMR	Difference of fit
Configural invariance	$\chi^2(58) = 125.65, p < .001$	0.981	0.971	0.999	0.036	0.028	–
Metric invariance	$\chi^2(64) = 155.78, p < .001$	0.975	0.964	0.999	0.040	0.033	$\chi^2(6) = 30.13, p < .001$
Scalar invariance	$\chi^2(70) = 189.45, p < .001$	0.967	0.957	0.998	0.044	0.038	$\chi^2(6) = 33.67, p < .001$
Residual invariance	$\chi^2(80) = 234.03, p < .001$	0.957	0.952	0.998	0.047	0.042	$\chi^2(10) = 44.58, p < .001$

Note. Confirmatory factor analysis based on a simple structure with subtests loading only on their respective latent factor, and freely estimated covariance between the four latent factors.

Lecerf et al., 2010) showed excellent fit in the WISC-IV standardization sample, $\chi^2(31) = 65.24, p < .001$, CFI = 0.988, TLI = 0.983, GFI = 0.988, RMSEA = 0.032, SRMR = 0.024, but due to the negative covariances involving the verbal comprehension latent variable, the same model flatly refused to converge in our gifted sample. This difference between the gifted sample and the standardization sample is compatible with the finding of two profiles of participants with opposite patterns on the verbal subtests vs. other subtests.

4. Discussion

The results of cluster analyses unambiguously revealed that referred children in the process of being identified as gifted with a Wechsler scale can show one of two distinct psychometric profiles, fully compatible with our predictions. The “homogeneous” profile has high performance

throughout all areas of cognitive ability assessed by the test (albeit to a lower extent for working memory and processing speed), and demonstrates a high FSIQ. By contrast, the “crystallized” profile is characterized by selectively very high performance on subtests of the VCI, with a large discrepancy with other indices, and a lower FSIQ on average. These two profiles are compatible with two major views of giftedness: one defined by high cognitive ability, and one defined primarily by high aptitude for acquiring and using declarative knowledge. In short, our results demonstrate the point repeatedly made by major authors in the field of giftedness (Gagné, 1995; Gallagher & Courtright, 1986) that the population of children labeled as “gifted” is not uniform.

Further analyses confirmed that the factor structure of the WISC in our gifted sample showed a negative correlation between the latent variable for VCI and other indices, in line with prior literature (Hagmann-von Arx et al., 2008; Liratny & Pry, 2012; Macmann et al.,

1991; Molinero et al., 2015; Pezzuti et al., 2022; Rowe et al., 2010, 2014; Watkins et al., 2002), but very different from typical results in non-gifted samples. This result departs even from the ability differentiation literature, which would not lead to predicting a negative correlation between indices. The two profiles were also absent from the standardization sample. This suggests that the two profiles found here are very specific to children referred for giftedness identification.

Our interpretation is that this largely stems from the existence of two major roads to being noticed and referred for giftedness identification, in countries where children are not systematically tested but are instead selectively referred by adults. Children may be referred as potentially gifted due to displaying superior cognitive ability (as manifested e.g. in rapid cognitive development, complex and unusual interests, ability to understand subtle forms of humor, etc.; Tourreix et al., 2023) or high ability to accumulate and use declarative knowledge (as manifested e.g. in high academic performance). As either motive is sufficient for referral and identification as gifted, most children in a referred sample end up leaning towards one profile or the other (although the separation is not clear-cut - see Fig. 3).

4.1. Alternative interpretations for the two profiles of performance

Our finding of two profiles in a referred sample, one with homogeneously high performance and one with high performance specifically on the VCI, is clear-cut and aligns with prior literature on ability profiles in gifted individuals (Lubinski, 2016) and ability differentiation (Blum & Holling, 2017; Breit, Brunner, et al., 2022). However, there are alternatives to our interpretation of these two profiles as representing children with high reasoning abilities overall vs. very high ability to acquire and use knowledge. Some, like the idea that the high VCI profile could be predominantly female (e.g. Kerr, 1997; see also Fishkin et al., 1994), can be ruled out from the data, but others require careful consideration.

One alternative is that the VCI-PRI difference reflects a contrast between “verbal” and “non-verbal” giftedness. Since VCI subtests conflate modality (verbal responses) with content (declarative knowledge – crystallized intelligence), it is difficult to conclusively work out the origin of high scores (see Schipolowski et al., 2014). Additional data (e.g. school grades) would be needed to rule out this possibility. However, VCI subtests appear particularly sensitive to culture-based declarative knowledge rather than verbal ability (e.g. Georgas et al., 2003; Gonthier et al., 2021; Watrin et al., 2023), the theoretical distinction between *Gf*/*Gc* is much more prevalent in intelligence research than the verbal vs. non-verbal dichotomy, and children in our dataset with high VCI scores tended to perform substantially lower on other tasks with a verbal modality such as Digit Span.

A second possibility is that children with a large discrepancy between indices are twice-exceptional, exhibiting both intellectual giftedness and a neurodevelopmental disorder (see e.g. Habib, 2023; Schiff et al., 1981; Volker et al., 2006), such as dyslexia ($VCI < PRI$) or dyspraxia/DCD ($VCI > PRI$). However, this seems highly unlikely given the frequency of the two profiles: certain neurodevelopmental disorders might be slightly more frequent in gifted samples (Toffalini et al., 2017), but it is very implausible that half of referred gifted children have a form of dyspraxia. Referral records also showed very little indication of learning disabilities. It seems much more likely that the high prevalence of profiles with elevated VCI misleads investigators into overestimating the frequency of learning disorders in gifted children.

A third possibility is that all children in the sample had high intelligence, with the two profiles differentiating those who successfully accumulated academic knowledge from those who did not (Gagné, 1985). A similar idea based on investment theory would be that in some gifted individuals, high fluid reasoning “snowballs” into high crystallized intelligence and an enhanced VCI (Rozenecwajg et al., 2009; see also Breit, Brunner, et al., 2022, for a similar mechanism as a possible explanation for the phenomenon of differentiation). However, our findings contradict this view: on average, children with a crystallized

profile had significantly lower scores on measures of logical reasoning, working memory and processing speed. In other words, about half of the referred gifted children exhibited superior crystallized intelligence despite fluid intelligence below the exceptional range.

4.2. Consequences for practice

Giftedness is an arbitrary social construct (Borland, 2005), broadly referring to individuals who demonstrate “exceptional abilities”, which can be defined in multiple ways (Sternberg & Davidson, 2005). As a result, there is no “correct” way to define giftedness, but our results make it clear that the different strategies currently in use – such as relying on intelligence tests vs. academic results (e.g. Cao et al., 2017; Carman, 2013; Ziegler & Raul, 2000), or using only FSIQ vs. labeling children as “gifted in a particular area” (e.g. Gridley et al., 2003; Silverman, 2009), will lead to meaningfully different groups of children.

Excluding children with a crystallized profile from the definition of “giftedness” seems unworthwhile, as they can achieve impressive results despite reasoning abilities in the non-exceptional range if offered the right educational opportunities.⁵ Identifying children based exclusively on the VCI (assuming that this is in any sense a “fairer” measure because many gifted children perform lower on the FSIQ and PRI: e.g. Silverman, 2009) is equally problematic, as it will both include children with a supportive environment for learning but non-exceptional reasoning ability, and overlook children with high cognitive abilities who could have succeeded if provided with different learning opportunities (e.g. Lupart & Pyryt, 1996). The problem is the same with two-stage identification methods using academic achievement as a first filter (Worrell & Erwin, 2011), or with considering that a child with high reasoning ability but low academic success “is no longer gifted in terms of the school’s curriculum” (Cross & Coleman, 2005).

Based on our results and the literature, we propose the following recommendations when using intelligence tests as part of a giftedness identification strategy:

1/ Make it explicit whether a given patient or sample has a homogeneous profile (FSIQ at least 130, or in the case of children with language difficulties, PRI or FRI/VSI at least 130, and overall high scores); or a crystallized profile (VCI at least 130 with lower scores on other indices), possibly using the rule of thumb of [average of indices other than the VCI] < [half the VCI + 50] (see the Results section). Note that the pattern for a given individual can be ambiguous, as recalled in Fig. 3; in this case, assigning a broad “gifted” label would be desirable.

2/ Carefully consider a gifted individual’s profile, as a way to better foster abilities (Kaufman & Sternberg, 2008; Rogers, 2007), predict areas of future success (Lubinski et al., 2001; Park et al., 2007; Webb et al., 2007), and tailor gifted-specific programs and recommendations (e.g. Lohman et al., 2008; Wai et al., 2022). For instance, one can imagine that a child with a homogeneous profile might be more likely to underachieve due to boredom and to experience a feeling of disconnection with their peers (Tourreix et al., 2023) and might benefit more from grade-skipping, whereas a child with a crystallized profile who is at ease in their current class might benefit more from course enrichment.

3/ Avoid identifying giftedness solely based on the VCI, or based on any exceptional score in a Wechsler scale (e.g. by identifying children as “processing speed gifted”, as is sometimes done in France, leading to much parent confusion and poor outcomes when these children are led to skip grades) as this will severely overestimate the prevalence of giftedness (see McBee & Makel, 2019; using “any index ≥ 130 ” as a criterion in the WISC will lead to about 10 % of gifted children). Instead,

⁵ A good illustration of this point is provided by Lewis Terman’s princeps longitudinal study of giftedness, which screened over 168.000 children to select 1528 gifted. Out of the initial 168.000 children, two ended up winning a Nobel prize; both famously failed to be selected for the study because their IQ was not high enough (see Warne et al., 2020).

carefully consider the FSIQ and expect children to lean towards one of the two profiles identified here.

4.3. Consequences for giftedness research

Studies on giftedness use very different criteria to identify participants. Some rely exclusively on teacher nominations, school grades, standardized academic exams (e.g. the SAT), or cognitive ability tests assessing a mix of school-trained abilities which in fact correlate almost perfectly with academic achievement (e.g. Roznowski et al., 2000, used a “general intelligence” test assessing vocabulary, reading and arithmetic which correlated $r = 0.87$ with SAT performance). This will primarily identify children with a crystallized profile. Others draw on performance on visuo-spatial logical reasoning tests such as Raven’s matrices, which will primarily identify children with a homogeneous profile. Studies using Wechsler-like scales will be sensitive to both profiles.

Despite using the same label of “giftedness” and being discussed interchangeably in the literature, these studies may yield conflicting conclusions. Research on fluid reasoning strategies (see e.g. Steiner & Carr, 2003) may not apply to children with a crystallized profile and non-exceptional reasoning ability, whereas psychosocial strengths or difficulties may differ by profile, which – combined with the use of varying criteria – could explain some of the major inconsistencies in the literature (e.g. Tourreix et al., 2023). As there can be no “correct” definition of giftedness, all strategies are valid but researchers must pay attention to the profile of giftedness their criteria identify. Both literature reviews and empirical studies should explicitly state to which profile(s) of giftedness they are applicable.

Our results also provide important context for prior studies regarding heterogeneity in gifted individuals. Our large-scale datasets make it clear that heterogeneity exists, but is not random: extreme discrepancies are characteristic of the crystallized profile. In other words, distinguishing between “homogeneous” and “heterogeneous” gifted (e.g. Hamdioui & Vaivre-Douret, 2022; Nusbaum et al., 2017; Suprano et al., 2019) will unwittingly confuse “heterogeneous” with “high knowledge / academic achievement” – although with the complication that this group may also include twice-exceptional children.

4.4. Consequences for research on intellectual assessment

The present results have indirect consequences for the debate regarding the usefulness of score profiles on intelligence tests. The approach of interpreting an individual’s pattern of strengths and weaknesses has been severely criticized on the grounds that most of the useful variance in an intelligence test resides at the level of the general factor of intelligence. In other words, scores at the index or subtest level would contain little information above and beyond FSIQ (Canivez et al., 2019; McGill & Busse, 2017; McGill & Dombrowski, 2019; Watkins et al., 2022; see also Carroll, 2000). This is in line with the typical results of cluster analyses with Wechsler scales, which tend to show that although there can be some limited usefulness to profile shape (Davison et al., 1996; Hale & Saxe, 1983; Watkins & Glutting, 2000), clusters are mostly driven by general profile elevation, not patterns of specific strengths and weaknesses (Hale & Saxe, 1983; McDermott, Glutting, Jones, & Noonan, 1989; McDermott, Glutting, Jones, Watkins & Kush, 1989; Moses & Pritchard, 1996; Watkins & Glutting, 2000).

Our results showing two clearly separable and meaningful profiles in the context of children referred for giftedness testing, along with prior literature regarding the long-term stability and predictive value of cognitive profiles in both the non-gifted (Coyle, 2018; Coyle et al., 2014) and the gifted (Achter et al., 1999; Benbow & Minor, 1990; Breit et al., 2021; Breit, Scherrer, & Preckel, 2022; Lohman et al., 2008; Lubinski et al., 2001; Park et al., 2007), provide a stark contrast with this skeptical conclusion. The inconsistency is, in fact, easy to reconcile: none of the studies criticizing the validity of interpreting profiles of strengths

and weaknesses in intelligence tests were conducted in a gifted sample (in fact, our own complementary analysis on the two WISC standardization samples also failed to identify meaningful profiles in non-gifted samples, in line with this literature). As discussed in the introduction, given the phenomenon of ability differentiation, scores are less correlated for individuals with high ability, which means the general factor has less weight for high-performing individuals. In this context, it makes sense that profiles of performance carry more meaning for individuals with high ability (see McGill, 2015; Wai et al., 2022).

Our results therefore highlight an important boundary condition for the conclusion that only the FSIQ should be interpreted, instead of profiles at the index level: this might be true in general, but not in samples characterized by differentiation – i.e., by a lower weight of general intelligence and more discrepancies between indices. This was the case for the current samples referred for giftedness testing, and this can also be expected to apply to other referred samples with major difficulties in one or more areas of cognitive ability (e.g. in the context of brain lesions or specific learning disorders).

4.5. Limitations and directions for future studies

Profile analyses with intelligence tests have been criticized in prior literature due to limited evidence for the predictive ability and stability of clusters (Carroll, 2000; Watkins, 2000). In other words, profile analytic studies have generally failed to demonstrate that the profile of an individual predicts useful variance beyond the test, and that individuals are assigned to the same cluster when retested, at a different point in time or with a different test (Loehlin, 2019). These two limitations also apply to our study: as we analyzed records retrospectively, there was no follow-up and no other data available for the patients in our sample. This was inevitable (this is the downside to analyzing a representative sample of referred children), but it would be useful in future studies to control that assignment to either WISC profile is actually stable and predictive of real-life outcomes. Given the discrepancy between findings of psychometric profiles in gifted and non-gifted samples, and given prior results showing that specific ability scores are stable in gifted individuals (Breit et al., 2021; Breit, Brunner, et al., 2022; Lohman et al., 2008) and predictive of actual achievement (Achter et al., 1999; Benbow & Minor, 1990; Lubinski et al., 2001; Park et al., 2007; Webb et al., 2007; as is also sometimes found in non-gifted samples, e.g. Coyle, 2018), we are confident that this will be the case.

Future research into profiles of giftedness should particularly investigate socio-economic or socio-cultural status as a possible correlate. High socio-economic status leads to both inflated crystallized intelligence scores, and discrepancies with other abilities (Seashore, 1951; Kaufman, 1976; for examples with the WISC, see Grégoire, 2009, 2021). This raises questions about whether the crystallized profile reflects genuine giftedness or family privilege (with the risk of offering more educational opportunities to already privileged children; see Carman & Taylor, 2010). This was difficult to assess in our sample, as socio-economic information was rarely recorded, and was uniformly high when it was (this is typical of giftedness assessment in French-speaking countries, where assessment is costly and privileged families are both more aware of giftedness and more likely to seek this type of social signaling).

Another factor insufficiently investigated in giftedness identification is the motive for referral (for examples, see Hannig & Koop, 2016). The reasons that lead to referral (in our sample, referral records showed that testing was frequently driven by exceptional school performance and a desire for grade-skipping, or conversely, boredom at school) are expected to be strongly related to a child’s profile. For instance, a crystallized profile may be more frequently referred due to exceptional results at school; this would be compatible with the slightly higher prevalence of the homogeneous profile in younger and female children (this profile may be more likely to be detected earlier, and possibly on the basis of signs other than academic achievement, although this could

also reflect a form of differentiation with age: Breit, Brunner, et al., 2022). However, this would need to be investigated directly.

4.6. Constraints on generality

Our results were obtained in a sample of referred children being tested for giftedness identification, and we expect the results to be specific to this type of population (rather than “the gifted” in general). Indeed, our interpretation is that the discrepancy between the two profiles reflects two possible roads to being noticed as possibly gifted and, therefore, referred for giftedness testing. This means the same two profiles are not necessarily expected to appear in a sample of unidentified gifted children (as could be recruited, for instance, by systematically screening a sample of the general population with an intelligence test): as testing is sought disproportionately by families with a high socio-cultural status which is also associated with higher crystallized intelligence, the Gc profile could be much less frequent in unidentified gifted children. It might also be the case that some children have different profiles altogether that do not lead to referral (e.g. very high working memory may be less noticeable), but the high correlation between indices other than the VCI makes it unlikely that other profiles are prevalent.

More generally, different practices in selecting which children are referred for testing will probably lead to different results regarding the frequency of the two profiles. For example, if all children included in a sample were referred based on teacher recommendations, it could be the case that most children display the crystallized profile. This could lead to different results in countries where referral is based on different criteria (see also Gonthier & Guignard, in press). However, the universality of large discrepancies in gifted samples (Gonthier, 2025) suggests that the two profiles of performance observed here may be relatively common.

In the case of countries where children are proposed for testing by parents or teachers based on any evidence of “exceptional abilities” in everyday life or at school, the results are expected to generalize well: the two profiles found here were robust to changes of analytic strategy, and appeared across multiple psychologists, two different countries, and two different versions of the WISC. Given that other intelligence tests also tend to include a mix of subtests assessing fluid and crystallized intelligence, and given that large heterogeneity in gifted samples has also been reported using the K-ABC II (Lichtenberger et al., 2006) and Woodcock-Johnson III (Rizza et al., 2001), we expect the results to generalize to other test batteries as well.

CRedit authorship contribution statement

Corentin Gonthier: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Data curation, Conceptualization. **Emma Toureix:** Writing – review & editing, Investigation, Data curation, Conceptualization. **Maud Besançon:** Writing – review & editing, Investigation, Conceptualization. **Jacques Grégoire:** Writing – review & editing, Investigation, Data curation. **Jacques-Henri Guignard:** Writing – review & editing, Investigation, Data curation.

Declaration of competing interest

The authors have no conflict of interest to report.

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Data availability

The data and materials for the study are not available (the data were all extracted from clinical records and the patients did not provide consent for their data to be made accessible to other researchers; the

Wechsler scales are commercially available).

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